

KEYSTRA ENERGY GROUP

SOLUTION 1:

ENTERPRISE HSE PERFORMANCE MANAGEMENT

Portfolio Project | Proof of Concept

Data Source: Synthetic (Python Engineered)

PROJECT OVERVIEW

Keystra Energy Group is a fictitious integrated energy company created for this portfolio, with operations across Upstream, Midstream, Downstream, LNG, and Renewables. This project develops an Enterprise HSE Performance Management solution to answer a practical business question:

How can HSE data be used to understand safety performance, identify areas requiring attention, and connect safety events to their operational and financial consequences?

The solution was designed around a simple principle: Incident counts alone are not enough.

Effective HSE performance management requires visibility into the amount of work being performed, the safety events occurring within that exposure, the consequences of those events, and the organisation's response. This is a portfolio proof of concept. It has not been deployed, implemented, or tested within a real company.

BUSINESS CHALLENGE

HSE information is generated across different operational processes. Workforce exposure records show the level of activity being performed. Incident records capture safety events. Consequence data captures their operational, workforce, and financial effects. Corrective-action records capture the organisation's response. When these areas are analysed separately, management can see individual metrics but may struggle to answer the more important questions:

- * Where is workforce exposure concentrated?
- * Are incident levels significant relative to that exposure?
- * Which hazards are producing the most serious outcomes?
- * Which assets are creating the greatest operational, workforce, and financial consequences?
- * Where does unresolved corrective-action risk remain?

The business problem is therefore not simply a lack of HSE data. It is the lack of an integrated analytical view connecting safety performance with business consequence.

SOLUTION DEVELOPED/DATA FOUNDATION

The project addresses this problem through an end-to-end analytical workflow:
Python Data Engineering → HSE Data Warehouse → HSE Analytics → Power BI

The analytical model connects:

Workforce → Exposure → HSE Incident → Consequence → Corrective Action

A synthetic enterprise HSE environment was engineered in Python to represent the organisational structure and operational relationships of an integrated energy company. The timeframe for this analysis spans between 1 January 2022 to 31 July 2026. The components of this data include 3 Facts table and 12 Dimension tables containing Business units, Departments, Assets, Incident types, Corrective action status and so on.

Analytics provided include:

- Workforce Exposure — measurement of workforce activity and exposure.
- Safety Performance — incident frequency, recordable events, TRIF and LTIFR.
- Consequence Analysis — lost workdays, downtime and direct incident cost.
- Hazard Analysis — identification of recurring and high-consequence hazard patterns.
- Corrective-Action Performance — assessment of open, overdue, verified and effective actions.
- Enterprise Reporting — Power BI reporting of HSE performance and findings.

KEY FINDINGS

The analysis identified five major findings from the simulated HSE environment.

1. *Workforce exposure was concentrated in the major operating businesses*

BUSINESS UNIT	EXPOSURE
Upstream	13.7M
Downstream	9.1M
Midstream	6.4M

This means that Workforce activity was therefore not evenly distributed across the enterprise.

This matters because incident volume must be interpreted against the amount of work being performed. An operation with more incidents may also have substantially greater workforce exposure. Exposure therefore provides the denominator needed to distinguish operational scale from underlying safety-performance signals.

2. *The enterprise showed both frequent safety events and high consequence outcomes*

The simulated environment recorded:

PERFORMANCE INDICATOR	RESULT
HSE Incidents	6,370
Recordable Incidents	702
Near Misses	3,710
TRIF	3.86
LTIFR	0.41
Fatalities	2

The dataset therefore contained both recurring lower-consequence events and less frequent but severe outcomes. This matters because frequency and consequence tell different stories.

Frequency helps identify recurring safety problems, while consequence shows where events are creating substantial workforce, operational, or financial impact. A complete HSE assessment therefore needs both perspectives.

3. Different assets emerged as priorities depending on the measure used

The analysis did not identify one universally “worst” asset. Three assets stood out for different areas of concern.

- Gas Processing Plant

This asset recorded the highest TRIF (6.02) amongst the assets analysed. The elevated injury frequency makes the asset a priority for investigating the conditions, activities, and controls contributing to the observed pattern.

- Offshore Platform A

This asset poses the greatest operational and financial consequence. It recorded the greatest combination of downtime, workforce impact, direct incident cost, corrective-action cost, and unresolved actions. The concentration of these consequences makes Offshore Platform A the strongest overall management signal in the analysis and warrants closer investigation of unresolved risks and control effectiveness.

- Refinery Complex

The Refinery Complex recorded the highest lost-workday burden among the assets analysed.

Medical Treatment Cases accounted for approximately 79.6% of lost workdays, with Vehicle/Mobile Equipment and Lifting & Material Handling accounting for almost all of that MTC lost-workday burden.

This demonstrates why incident frequency alone can produce an incomplete picture. The scale of workforce impact warrants further investigation into the activities, hazards, and controls associated with these incidents.

4. Recurring hazards and high consequence events require different types of attention

The following hazard categories repeatedly appeared in the analysis: Vehicle/Mobile Equipment, Lifting & Material Handling, Ergonomic/Manual Handling, Hydrocarbon Release, and Dropped Objects.

Key observations include:

- * Vehicle/Mobile Equipment and Lifting & Material Handling were prominent in recordable injuries and workforce consequences.
- * Hydrocarbon Release was an important process-safety signal at Offshore Platform A, with 32 incidents and continued corrective-action concerns.
- * A Dropped Object fatality at Offshore Platform B demonstrated that lower-frequency hazards can still produce severe outcomes.

Hazard prioritisation should therefore consider both frequency and consequence. Recurring hazards may indicate persistent control weaknesses, while lower-frequency hazards may require attention because of their potential or actual severity. Frequency identifies recurring problems; consequence identifies where those problems matter most.

5. *Corrective-Action closure does not necessarily indicate risk reduction*

The solution analysed 3,060 corrective actions across the enterprise.

At Offshore Platform A, important actions remained:

* Open * Overdue * Partially effective * Not yet verified

A corrective action being marked Closed does not necessarily demonstrate that the underlying risk has been reduced. A stronger monitoring approach should consider:

Priority → Timeliness → Closure → Verification → Effectiveness

This shifts corrective-action management from simply tracking completion to assessing whether the action has actually strengthened the relevant control.

OVERALL ANALYTICAL CONCLUSION

The central finding is that enterprise HSE performance cannot be represented by incident counts alone. The analysis shows that:

- Exposure explains the scale of activity.
- Safety frequency identifies recurring events.
- Consequence identifies workforce, operational, and financial impact.
- Hazard analysis identifies recurring and high-consequence signals.
- Corrective-action analysis identifies where the organisational response may remain incomplete.

Together, these perspectives provide a more complete view of HSE performance and support more targeted investigation and prioritisation.

PRIORITIES AND RECOMMENDATIONS

Based on the findings, the proof-of-concept highlights five areas for management attention:

PRIORITY	ANALYTICAL SIGNAL	PROPOSED FOCUS
Gas Processing Plant	Highest TRIF	Investigate drivers of elevated injury frequency
Offshore Platform A	Greatest downtime, cost, and unresolved actions	Strengthen control assurance and address high priority outstanding risks
Refinery Complex	Highest Lost workday burden	Investigate vehicle/mobile equipment and material handling exposure and controls

Across assets	Recurring vehicle/mobile equipment patterns	Conduct cross-asset review for common control weaknesses
High consequence hazards	Hydrocarbon Release / Dropped Objects	Strengthen preventive and critical-control assurance

These are analytical recommendations generated from the proof of concept, not actions implemented within a real Keystra organisation.

BUSINESS VALUE

The solution demonstrates a shift from reactive incident reporting towards integrated HSE performance management. It enables the organisation to move from:

“How many incidents occurred?”

to:

“Where is exposure concentrated, what safety outcomes are occurring, what consequences are being created, and where does unresolved risk remain?”

SCOPE & LIMITATIONS

Keystra Energy Group is a fictitious organisation created for this portfolio. The HSE dataset is synthetic and was engineered in Python specifically for this project.

The solution is a proof of concept not a real company deployment, therefore the results represent patterns within the simulated environment.

The analysis identifies relationships and signals but does not establish causation. The proposed priorities would require validation against real operational data, field observations, HSE investigations, and subject-matter expertise before implementation.

The numerical results demonstrate the analytical approach, rather than predicting actual incident rates, safety outcomes, or business losses.

PROJECT ARTEFACTS & FULL ACCESS

Available Artefacts:

Full Project Report | Executive Summary | Power BI Report | Dashboard Pictures | Python Code | Data Warehouse

These are all available through the link below

Github:

https://github.com/winnerEM/Keystra-Energy-Solution-1_Enterprise-HSE-Performance-Management

Also accessible on

Google drive:

<https://drive.google.com/drive/folders/1Yr93K8r1ybdGdX4TRtibGDDTO4vx8bpu?usp=sharing>